X-Ray Fluorescence Analysis

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X-ray fluorescence (XRF) spectra excited at 35 kV were used to identify the elements in six samples and to estimate the composition of a leaded brass. After a two-point energy calibration, fitted line centroids agreed with tabulated energies to within 0.05 keV (0.07 keV for the weakest line), small compared with the measured detector resolution of 0.46 keV (full width at half maximum at 6.4 keV). The brass contained Cu, Zn and Pb, and the magnetic target Sm and Co. The old 50-cent coin is a Cu-rich Cu–Ni alloy, while the new 50-cent and 1-dollar coins are Ni-rich with Cu; the new 50-cent coin lacks the Cu K_β line, consistent with Cu beneath a Ni surface layer that absorbs it. Refitting all brass and reference spectra with one three-Gaussian model and applying the manual's first-order mass-ratio relation gave Cu (60 ± 3)%, Zn (37 ± 2)% and Pb (2.5 ± 0.9)% by mass, consistent with free-cutting leaded brass; matrix effects are not included. The total count rate fell by 27–48% during every 900 s acquisition. This drift cancels within each brass spectrum but not between the brass and reference spectra, and the resulting scatter of the reference heights, not counting statistics, dominates the uncertainty.

1. INTRODUCTION

The elements in a material, and their proportions, govern much of its behaviour: the corrosion resistance of a coin, the machinability of a brass fitting and the strength of a permanent magnet all depend on composition. X-ray fluorescence (XRF) measures composition without damaging the sample. A primary X-ray photon ejects a core electron, an electron from a higher shell fills the vacancy, and the energy difference is emitted as a characteristic photon whose energy is set by the atomic number. Because the inner shells take no part in chemical bonding, these energies identify an element regardless of its chemical state, and the spectrum of a mixture is, to first order, the sum of the spectra of its components [1].

The experiment had two parts. Part 1 identified the elements in a brass target, a magnetic target and four Singapore coins from their line energies and line-family patterns. Part 2 estimated the mass fractions of Cu, Zn and Pb in the brass from fitted peak heights relative to pure-element references, with repeated measurements to assess reproducibility.

2. THEORY

2.1 Characteristic X-ray Emission

When a K-shell vacancy is filled from the L shell, a K_α photon is emitted; filling from the M shell gives K_β, which has the higher energy because the M–K gap is larger (Fig. 1). The 20 eV splitting of the 2p level divides K_α into $K_{\alpha1}$ and $K_{\alpha2}$, which the detector does not resolve. K_β is the less probable transition, and its intensity is about one fifth to one tenth of K_α; in the L series, L_α is accompanied by L_β of similar intensity and a weaker L_γ [1]. These patterns serve as identification checks. Line energies increase monotonically with atomic number (Moseley's law) [3], so the K_α lines of adjacent elements are only about 0.6 keV apart, and lines of different series can nearly coincide: Ni K_β (8.265 keV) and Cu K_α (8.044 keV) are 0.22 keV apart, and Cu K_β (8.905 keV) and Zn K_α (8.638 keV) 0.27 keV apart [2].

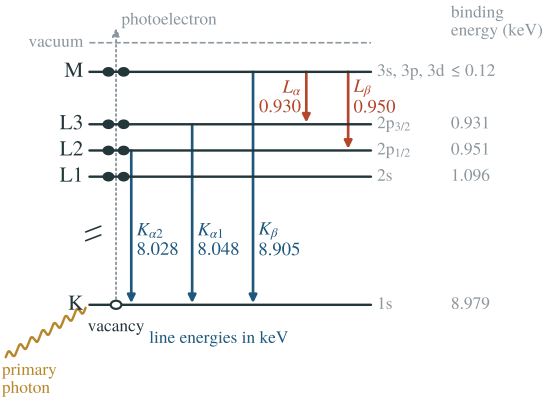


FIG. 1. Characteristic emission of Cu (energy axis broken and not to scale). A primary photon ejects a K electron, leaving a vacancy. An electron from the L_2 or L_3 subshell filling it emits $K_{\alpha2}$ or $K_{\alpha1}$, and an M electron emits K_β; each line energy is the difference of the two binding energies [2]. An M electron filling an L vacancy emits L_α or L_β.

An element absorbs photons strongly just above its K absorption edge and much less just below it. The Ni K edge (8.333 keV) lies between Cu K_α and Cu K_β, so Ni attenuates Cu K_β far more than Cu K_α [3]. For Pb the K edge is at 88.0 keV [3], above the maximum photon energy of a 35 kV tube, so only its L lines can be excited: L_α (10.551 keV), L_β (12.614 keV) and L_γ (14.764 keV) [2].

2.2 First-Order Quantitative Model

The manual's model takes the height of a peak as proportional to the number of emitting atoms in the irradiated volume [1]. Consider the irradiated area S and effective depth d, the same in

the alloy and in a pure reference of each element i, with density ρ_i and atomic mass A_i. The steps are:

$$\text{reference:}\quad n_{0,i} = \frac{Sd\rho_i}{A_i}$$

$$\text{alloy:}\quad n_i = n_{0,i}\frac{H_i}{H_{0,i}} = \frac{Sd\rho_i}{A_i}\cdot\frac{H_i}{H_{0,i}}$$

$$\text{mass fraction:}\quad C_i = \frac{n_i A_i}{\sum_j n_j A_j} = \frac{Sd\rho_i H_i \,/\, H_{0,i}}{\sum_j Sd\rho_j H_j \,/\, H_{0,j}},$$

where n is the number of emitting atoms and H_i, $H_{0,i}$ are the heights of the same line in the alloy and in the reference. Cancelling Sd gives

$$C_i = \frac{\rho_i H_i \,/\, H_{0,i}}{\sum_j \rho_j H_j \,/\, H_{0,j}}, \tag{1}$$

in which neither Sd nor A_i appears [1]. The ratio $H_i \,/\, H_{0,i}$ is the volume fraction of element i, so these ratios should sum to one, and the factor ρ_i converts volume to mass. An intensity factor common to all lines of one alloy spectrum cancels in Eq. (1); a factor that differs between the alloy spectrum and a reference spectrum does not. The model neglects matrix effects: absorption of the primary and fluorescent radiation by the other elements, and secondary fluorescence [1,4].

2.3 Gaussian Deconvolution

In brass, Cu K_β lies under the high-energy side of Zn K_α, so the Zn height cannot be read directly from the spectrum. The manual resolves the 7.5–9.1 keV region with three Gaussians of equal width at the known energies of Cu K_α, Zn K_α and Cu K_β [1]. The model fitted here is

$$N(E) = b + \sum_k A_k \exp\!\left[-(E - E_k - \delta)^2 \,/\, (2\sigma^2)\right], \tag{2}$$

where N is the number of counts per channel, A_k the height of line k, E_k its tabulated energy [2], σ the common width, b a constant background and δ a common energy offset. The terms b and δ are additions to the manual's model; Sec. 5.1 gives the reason.

3. APPARATUS

The spectrometer (Fig. 2) follows the manual [1]. An X-ray tube with a Mo anode, run at 35 kV and 1 mA, emits a primary beam that a collimator narrows onto the target. The target sits on a rotatable table set to 45° to the beam, so that the incident and detected directions make equal angles with its surface. A Si semiconductor detector, set at 90° to the beam, produces for each absorbed photon a pulse whose height is proportional to the photon energy. The MCA box sorts the pulses into 512 channels by height, and the CASSY Lab 2 software displays the resulting spectrum, applies the energy calibration and fits Gaussians to selected peaks. Placing the detector at 90° keeps it out of the primary beam, so the spectrum contains mainly

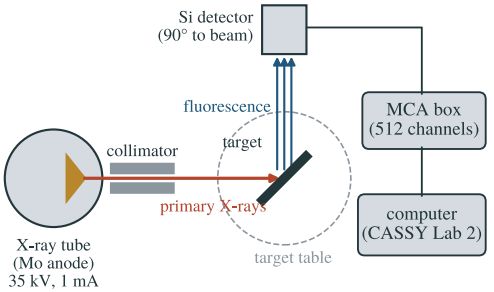


FIG. 2. The X-ray fluorescence spectrometer, viewed from above. The collimated primary beam strikes the target at 45°; the detector, at 90° to the beam, records the fluorescence, and the multichannel analyser (MCA) sorts its pulses by energy.

fluorescence and the part of the primary beam scattered by the target.

4. PART 1: QUALITATIVE ELEMENTAL IDENTIFICATION

4.1 Methods

The spectrometer was set up as in Fig. 2, with the settings of Table VI in the Appendix. Every spectrum was acquired for 900 s; the manual suggests 300 s. The energy scale was calibrated with the Fe+Zn target: the software located the Fe K_α and Zn K_α peak centres at channels 165.85 and 224.24, which were assigned 6.40 and 8.64 keV, giving 0.0384 keV per channel. This calibration was applied to every later spectrum. The brass (target 3), the magnetic sample (target 4) and the old and new 50-cent and 1-dollar coins were then measured.

Lines were identified by matching energies to tabulated values [2] and checking the line-family patterns of Sec. 2.1. Centroids, widths and heights were obtained by least-squares fits of Gaussians with a constant background, weighted by Poisson counting errors; quoted fit uncertainties are standard errors from the fit covariance.

4.2 Results

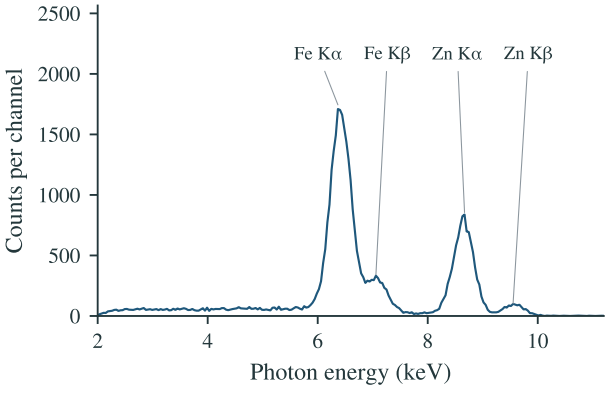


FIG. 3. Spectrum of the Fe+Zn calibration target, 900 s. Here and in Fig. 4, Fig. 5 and Fig. 6, each label is joined to the highest channel within 0.2 keV of the tabulated line energy; the range below 2 keV, which holds fewer than 0.3% of the counts, is not shown.

The calibration spectrum (Fig. 3) contains the K_α and K_β lines of Fe and Zn above a low continuum between 2 and 7 keV, present in every spectrum. Fewer than 0.3% of the counts in any spectrum were recorded below 2 keV, so lines below that energy, such as Al K_α (1.49 keV), could not be detected.

The detector resolution sets which lines can be separated. The fitted full width at half maximum (FWHM) of Fe K_α is (0.462 ± 0.004) keV; Zn K_α gives (0.433 ± 0.004) keV, Cu K_α in brass (0.444 ± 0.006) keV and Co K_α (0.474 ± 0.005) keV. Lines closer than about 0.45 keV therefore merge: the Ni K_β/Cu K_α (0.22 keV) and Cu K_β/Zn K_α (0.27 keV) pairs each appear as one peak, and their components can be separated only by fitting.

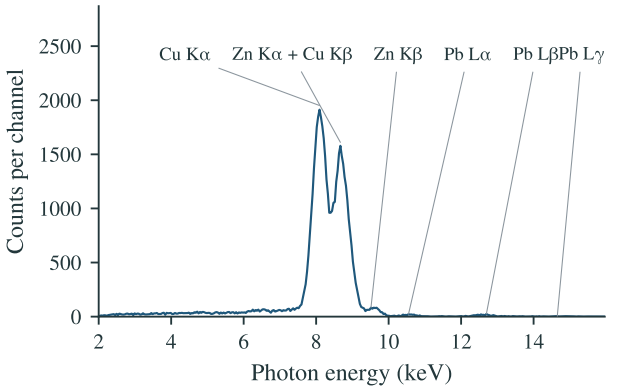


FIG. 4. Brass (target 3) spectrum, 900 s. The peak near 8.7 keV is the unresolved sum of Zn K_α and Cu K_β. The Pb L lines are about 16 counts per channel high (L_γ: 3) and are barely visible on this scale; their fitted positions are in Table I.

The brass spectrum (Fig. 4) contains Cu K_α, a second peak formed by Zn K_α and Cu K_β, Zn K_β, and the Pb L lines. The Pb lines are weak, but all three lie at their tabulated energies with the L-series intensity pattern, which identifies Pb.

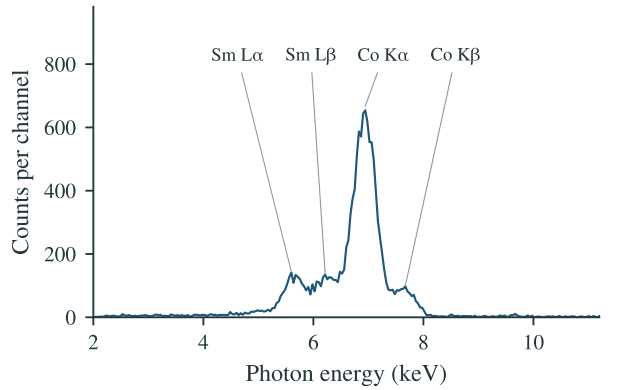


FIG. 5. Spectrum of the magnetic sample (target 4), 900 s, showing the Co K lines and the Sm L lines.

The magnetic sample (Fig. 5) shows Co K_α and K_β and Sm L_α and L_β, consistent with a samarium–cobalt permanent magnet. The Sm L_β centroid lies 0.04 keV above the tabulated L_{β_1} energy, consistent with a contribution from the unresolved L_{β_2} line at 6.587 keV [2].

TABLE I. Fitted line centroids compared with tabulated energies [2]. Uncertainties are fit standard errors. The Sm L_β entry is compared with L_{β_1}.

Line	Tabulated (keV)	Fitted (keV)	Δ (keV)
Calibration target			
Fe K_α	6.403	6.413 ± 0.002	+0.010
Fe K_β	7.057	7.080 ± 0.007	+0.023
Zn K_α	8.638	8.647 ± 0.002	+0.009
Zn K_β	9.572	9.561 ± 0.007	−0.011
Brass (target 3)			
Cu K_α	8.044	8.089 ± 0.002	+0.045
Pb L_α	10.551	10.567 ± 0.017	+0.016
Pb L_β	12.614	12.605 ± 0.018	−0.009
Pb L_γ	14.764	14.83 ± 0.05	+0.07
Magnetic sample (target 4)			
Sm L_α	5.636	5.639 ± 0.008	+0.003
Sm L_β	6.205	6.247 ± 0.010	+0.042
Co K_α	6.930	6.928 ± 0.003	−0.002
Co K_β	7.649	7.635 ± 0.010	−0.014

Table I compares fitted centroids with tabulated energies. In the calibration spectrum all four lines lie within 0.023 keV of their tabulated values. The brass and target-4 spectra deviate by up to 0.045 keV, apart from the weak Pb L_γ line ($+0.07 \pm 0.05$ keV). The Part 2 fits find a common offset of +0.013 to +0.039 keV in the brass spectra (Sec. 5.1), so the largest deviations are consistent with a small shift of the energy scale after calibration. Every deviation is below one sixth of the FWHM and well below the 0.6 keV spacing of adjacent elements, so none affects an identification.

The line-family patterns agree with the manual's description. The K_β/K_α height ratios are 0.173 ± 0.007 (Fe), 0.11 ± 0.01 (Zn), 0.126 ± 0.007 (Co) and 0.122 ± 0.007 (pure Cu reference), all between 1/10 and 1/5. The L-series ratios are Sm $L_\beta/L_\alpha = 0.99 \pm 0.06$, Pb $L_\beta/L_\alpha = 1.0 \pm 0.2$ and Pb $L_\gamma/L_\alpha = 0.18 \pm 0.05$. These uncertainties are scaled by $\sqrt{\chi^2_\nu}$, where χ^2_ν is the reduced chi-squared of the fit.

The coins (Fig. 6) contain Ni and Cu in different proportions. Because Ni K_β and Cu K_α are unresolved, the four Ni and Cu lines were fitted together, and Table II gives two height ratios. A pure Cu reference spectrum fitted in the same way serves as a control: its apparent Ni K_α/Cu K_α ratio of 0.015 measures the low-energy tail of the Cu K_α line, which a Gaussian does not describe. The same line-shape mismatch gives $\chi^2_\nu = 4$–9 for the coins.

TABLE II. Height ratios from fits of Ni K_α, Ni K_β, Cu K_α and Cu K_β (fixed tabulated spacings, common width and offset, constant background, 6.9–9.5 keV). Uncertainties are scaled by $\sqrt{\chi^2_\nu}$.

Spectrum	Ni K_α/Cu K_α	Cu K_β/Cu K_α	χ^2_ν
Pure Cu (control)	0.015 ± 0.004	0.122 ± 0.007	4.0
Old 1 dollar	0.047 ± 0.005	0.117 ± 0.006	4.3
Old 50 cents	0.32 ± 0.01	0.060 ± 0.006	5.4
New 50 cents	2.3 ± 0.1	0.005 ± 0.005	6.9
New 1 dollar	2.3 ± 0.2	0.19 ± 0.02	8.9

In the old 50-cent coin, Cu K_α dominates and Ni K_α appears as a resolved shoulder (Ni/Cu = 0.32), consistent with a Cu–Ni alloy. The Cu K_β/K_α ratio is half that of pure Cu, consistent

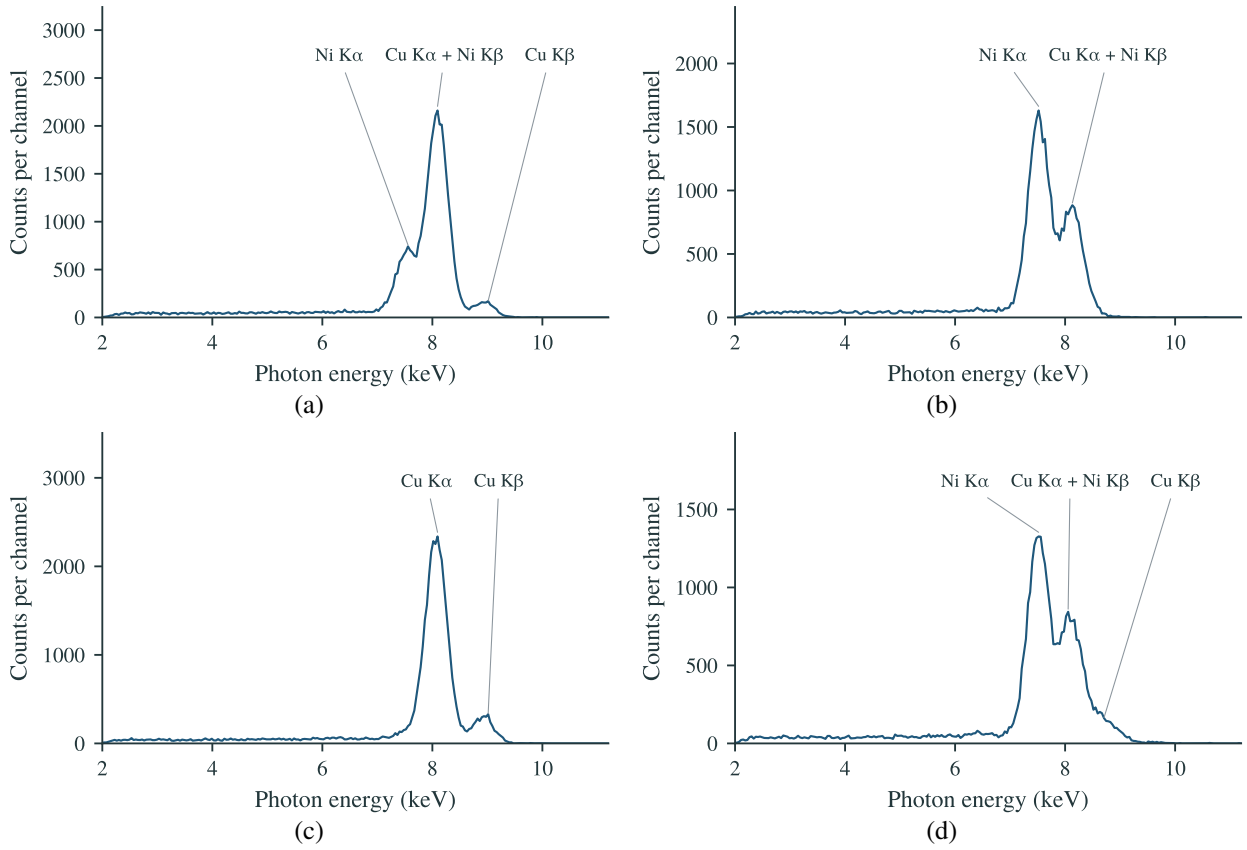


FIG. 6. Spectra of the Singapore coins, 900 s each, drawn as in Fig. 3: (a) old 50-cent, (b) new 50-cent, (c) old 1-dollar, (d) new 1-dollar. In (a), (b) and (d) Cu K_α and Ni K_β form one peak near 8.1 keV. The new 50-cent coin (b) has no Cu K_β peak.

with absorption of Cu K_β by Ni distributed through the alloy (Sec. 2.1).

In the new 50-cent coin, Ni K_α is the strongest line (Ni/Cu = 2.3) and Cu K_α is clearly present, but Cu K_β is absent (0.005 ± 0.005 of Cu K_α, against 0.122 for pure Cu). Cu atoms that emit K_α also emit K_β, so the missing line is consistent with Cu beneath a Ni surface layer that absorbs Cu K_β and transmits part of Cu K_α.

The new 1-dollar coin is also Ni-rich with Cu (Ni/Cu = 2.3). A shoulder between 8.5 and 9.1 keV is not described by the Ni and Cu lines alone: adding a Zn K_α component lowers χ^2_ν from 8.9 to 4.7. A Zn contribution is therefore possible but not established, and the Cu K_β/K_α value for this coin is unreliable.

In the old 1-dollar coin, Cu lines dominate, with the K_β/K_α ratio of pure Cu. A weak component at the Ni K_α energy (0.047 of Cu K_α) is three times the tail level of the control, so a small Ni content is possible, but the imperfect line shape prevents a firm identification. An alloying element with lines below 2 keV, such as Al, would not be detected.

4.3 Discussion

The new coins were attracted to a magnet and the old coins were not, which points to a ferromagnetic core in the new coins. No Fe lines appear in their spectra. XRF samples only the surface layer from which fluorescent photons escape, so Fe K_α photons from a steel core beneath a sufficiently thick Ni and Cu plating would not reach the detector; the absence of Fe therefore does not exclude a steel core. The plating thickness was not measured, so this cannot be tested quantitatively. The layered structure inferred for the new 50-cent coin from its missing Cu K_β line is consistent with such a plated construction.

5. PART 2: QUANTITATIVE BRASS COMPOSITION

5.1 Methods

The brass was measured five times and each pure reference (Cu, Zn, Pb) four times, for 900 s each, with the sample repositioned between runs. The acquisition software's Gaussian fits were not used for the final heights, because they were not uniform: three of the five brass fits included a fourth Gaussian at 9.18 keV, with negative height in two, and the fit windows differed between runs. All 17 spectra were instead refitted with Eq. (2): three Gaussians over 7.5–9.1 keV and one over 10.0–11.5 keV (Pb L_α) for the brass, as in the manual [1], and one Gaussian over the whole K_α or L_α peak for each reference (Cu 7.5–8.6 keV, Zn 8.1–9.2 keV, Pb 10.0–11.5 keV).

The offset δ was needed for an adequate fit. With the centres fixed at the tabulated energies, χ^2_ν of the brass fits was 2.7–12; with δ free it was 0.7–1.4, for $\delta = +0.013$ to $+0.039$ keV. With δ and b free, χ^2_ν was 1.3–2.1 for Pb L_α in the brass and 0.9–2.6 for the references.

Eq. (1) was evaluated with the mean heights, and its uncertainty was propagated by Monte Carlo with 10^5 draws. The mean brass heights were drawn from a multivariate normal distribution with the covariance of the five trials divided by five, which keeps the correlation between the three heights of one spectrum. Each mean reference height was drawn from a log-normal distribution whose relative width is the single-run relative standard deviation, not the standard error, because the reference intensities drifted during each series of runs (Sec. 5.3.1) and the runs are not independent samples of one level. The quoted uncertainty is the standard deviation of the resulting mass fractions.

5.2 Results

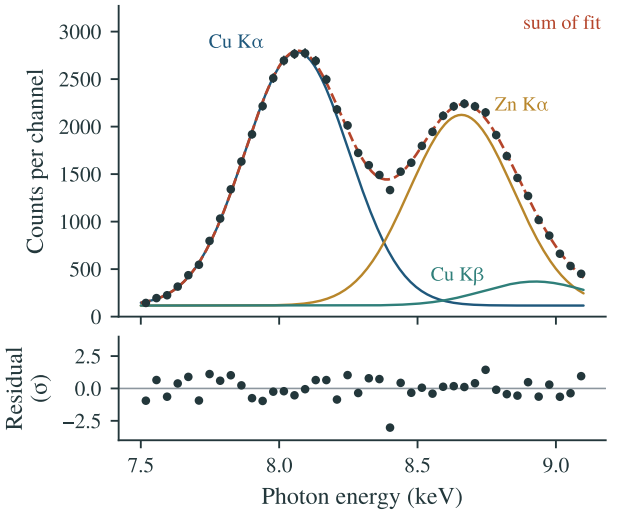


FIG. 7. Top: brass spectrum of trial 1 (points, Poisson error bars) with the fitted Cu K_α, Zn K_α and Cu K_β Gaussians, each drawn on the fitted background, and their sum (dashed). Bottom: residuals in units of the Poisson error.

The three-Gaussian model describes the overlap region within counting noise. Fig. 7 shows the fit to the most intense brass spectrum: $\chi^2_\nu = 0.74$ for 36 degrees of freedom, with no systematic structure in the residuals. The fitted heights are listed in Table III and Table IV.

TABLE III. Fitted brass heights (counts per channel) with fit standard errors.

Trial	H_{Cu}	H_{Zn}	H_{Pb}
1	2668 ± 21	2007 ± 19	26.6 ± 1.8
2	1954 ± 17	1503 ± 16	21.4 ± 1.7
3	1782 ± 17	1310 ± 16	20.0 ± 1.9
4	2428 ± 20	1877 ± 19	20.6 ± 1.5
5	1709 ± 16	1280 ± 15	17.7 ± 1.9
Mean ± s.d.	2108 ± 420	1595 ± 331	21.3 ± 3.3

TABLE IV. Fitted reference heights H_0 (counts per channel) in chronological order. Fit standard errors are 8–17 counts, at most 2%.

Line	Run 1	Run 2	Run 3	Run 4	Mean ± s.d.
Cu K_α	2013	1905	1740	1687	1836 ± 150
Zn K_α	1875	1907	1711	1654	1787 ± 123
Pb L_α	834	563	460	443	575 ± 181

TABLE V. Mass fractions of the brass from Eq. (1) with the mean heights of Table III and Table IV. Uncertainties are Monte Carlo standard deviations and exclude matrix effects.

Element	ρ (g/cm³)	Line	H/H_0	$\rho H/H_0$	C_i (%)
Cu	8.96	K_α	1.148	10.29	60 ± 3
Zn	7.14	K_α	0.893	6.37	37 ± 2
Pb	11.34	L_α	0.0370	0.420	2.5 ± 0.9

The brass composition is Cu (60 ± 3)%, Zn (37 ± 2)% and Pb (2.5 ± 0.9)% by mass (Table V). The central values are 60.2%, 37.3% and 2.5%, and the 16th–84th percentile ranges of the Monte Carlo distributions are 57.6–62.6%, 34.8–39.7% and 1.9–3.5%.

5.3 Analysis

Three effects bear on the composition: the drift of the detected intensity during and between runs, the choice of fit model, and counting statistics.

5.3.1 Intensity Drift

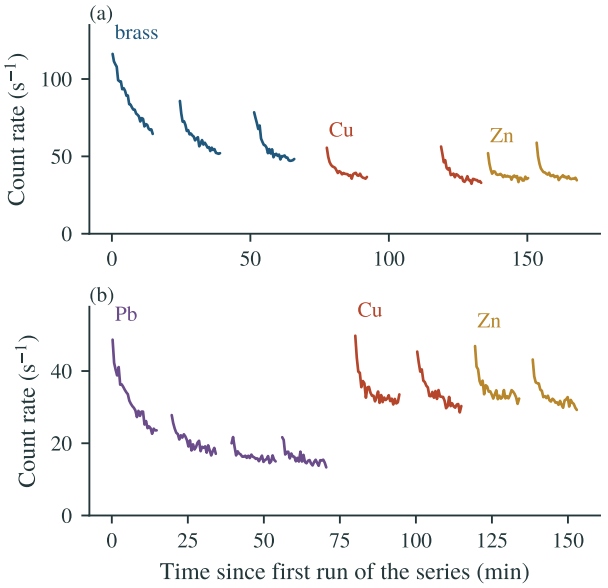


FIG. 8. Total count rate (30 s averages) during the two series of Part 2 runs, (a) and (b), against time since the first run of each series. Each segment is one 900 s run; gaps are sample changes.

The heights varied far more than counting statistics allow, because the detected intensity changed during and between runs. The brass Cu K_α height ranged from 1709 to 2668 with a 1% fit uncertainty, and the Pb reference height fell from 834 to 443 over four consecutive runs. Fig. 8 shows the total count rate, which fell by 27–48% between the first and last minute of each of the 17 runs and recovered only partly between runs. The reference heights followed this overall intensity: the height per total count varied by at most 5% for each reference, while the heights themselves varied by up to a factor of 1.9. The decay within a run cannot come from repositioning. It is consistent with a warm-up transient of the tube or detector after the high voltage is switched on, as the manual instructs for each run, but the cause was not identified.

The drift cancels within a brass spectrum but not between spectra. The three heights of one brass spectrum share the same intensity factor, which cancels in Eq. (1): the Cu/Zn height ratio varied only from 1.29 to 1.36 while the heights changed by a factor of 1.6, and mass fractions computed for each trial separately span 59.7–60.7% (Cu), 36.5–38.0% (Zn) and 2.1–2.8% (Pb). The reference heights share no factor with the brass or with each other, so their scatter (relative standard deviations of 8% for Cu, 7% for Zn and 31% for Pb) enters the result directly. The Monte Carlo attributes ±2.5, ±2.4 and ±0.8 percentage points (Cu, Zn, Pb) to the references and only ±0.2, ±0.2 and ±0.1 to the brass trials.

A common intensity factor also explains why $H/H_0 = 1.15$ for Cu, although the alloy contains less Cu per unit volume than the pure reference. The volume fractions H/H_0 sum to 2.08 instead of one, so the brass spectra were about twice as intense, relative to the references, as the model expects. A factor shared by the three brass lines, for example a larger irradiated area of the brass target, would produce this, and it cancels in Eq. (1).

5.3.2 Fit Model and Counting Statistics

The choice of fit model changes the result by at most 0.7 percentage points. Fixed centres without a background, as in the acquisition software, give Cu 60.9%, Zn 36.8% and Pb 2.3%; fixed centres with a background give 60.5%, 37.1% and 2.4%; the heights recorded by the software give 60.8%, 36.6% and 2.6%.

Counting statistics limit only the Pb measurement. The fit uncertainty of the brass Pb L_α height was 8% per 900 s trial, against 0.9% for Cu K_α and 1.1% for Zn K_α. A fit uncertainty from Poisson counts scales as $t^{-1/2}$ with acquisition time t, so it would be 10% at 600 s, 14% at 300 s and 20% at 150 s for Pb, but only 2% at 150 s for Cu. The 11% scatter of the per-trial Pb fraction matches the 8% fit uncertainty, so the brass Pb height is counting-limited. The 31% scatter of the Pb reference is not, and longer runs would not reduce it. A significant background adds its own variance to that of the net peak, $\sigma^2_{\mathrm{net}} = \sigma^2_{\mathrm{peak}} + \sigma^2_{\mathrm{bkg}}$; here b is a fit parameter, so its uncertainty is included in the fit covariance. Under the 20-count Pb L_α peak, b was 1.4–2.4 counts per channel.

5.4 Discussion

The composition is consistent with free-cutting leaded brass (UNS C36000, nominally 61.5% Cu, 35.5% Zn and 3% Pb), each fraction agreeing within its uncertainty. The uncertainty excludes the matrix effects that Eq. (1) neglects [1,4]. Zn K_β (9.572 keV) lies above the Cu K edge (8.979 keV) and can excite Cu fluorescence in the alloy that is absent in the pure Cu reference, raising H_{Cu}; the Cu lines lie below the Zn K edge (9.659 keV) and cannot excite Zn [2]. Each line is also attenuated by a different matrix in the alloy and in its reference. These effects were not quantified; quantitative XRF corrects them with fundamental-parameter or empirical-coefficient methods [4,5]. The detector efficiency cancels in H/H_0, because both heights belong to the same line.

6. CONCLUSION

Fitted line centroids agreed with tabulated energies to within 0.05 keV (0.07 keV for the weak Pb L_γ line), well below the measured detector resolution of 0.46 keV FWHM at 6.4 keV. They identified Cu, Zn and Pb in the brass, Sm and Co in the magnetic target, Cu with Ni in the old 50-cent coin, Ni with Cu in the new 50-cent and 1-dollar coins, and Cu in the old 1-dollar coin. The missing Cu K_β line of the new 50-cent coin is consistent with Cu beneath a Ni surface layer. With the manual's first-order model, the brass contains Cu (60 ± 3)%, Zn (37 ± 2)% and Pb (2.5 ± 0.9)% by mass, consistent with free-cutting leaded brass; matrix effects are not included.

The count rate fell by 27–48% within every run, so the reference heights scattered by 7–31% and set almost the whole composition uncertainty. Specific improvements follow from the limitations found. Keeping the high voltage on and starting each acquisition only once the count rate has stabilised would remove the decay of Fig. 8. Interleaving the brass and reference measurements within one series, with a reference repeated at its start and end, would let any remaining drift be measured and divided out of H/H_0. Longer brass acquisitions would reduce the 8% counting uncertainty of the Pb height per 900 s run. A detector line shape that includes the low-energy tail would remove the apparent Ni in the pure Cu control and improve the coin fits ($\chi^2_\nu = 4$–9). A fundamental-parameter matrix correction would remove the bias of the first-order model, which the quoted uncertainty does not include.

APPENDIX

TABLE VI. Spectrometer settings.

Parameter	Value
Tube anode	Mo
Tube voltage, emission current	35 kV, 1 mA
Target angle, detector angle	45°, 90°
Detector	Si semiconductor
MCA	512 channels, input factor −5
Acquisition time	900 s per spectrum
Calibration target	Fe + Zn
Calibration points	6.40 keV at ch. 165.85, 8.64 keV at ch. 224.24
Energy per channel	0.0384 keV

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